

A close-up photograph of a dental model showing several teeth with white ceramic crowns. The crowns are fitted over the natural teeth, and a red line is visible on the gingival margin of each crown, indicating the preparation site. The background is dark blue.

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Dental Ceramics III

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Date : 11/ 4 /2024

Continue mechanical properties

B- Flexural strength (Craig Textbook 2020)

Tetragonal zirconia (3Y-TZP) ceramics exhibit the highest values (900 to 1500 MPa),

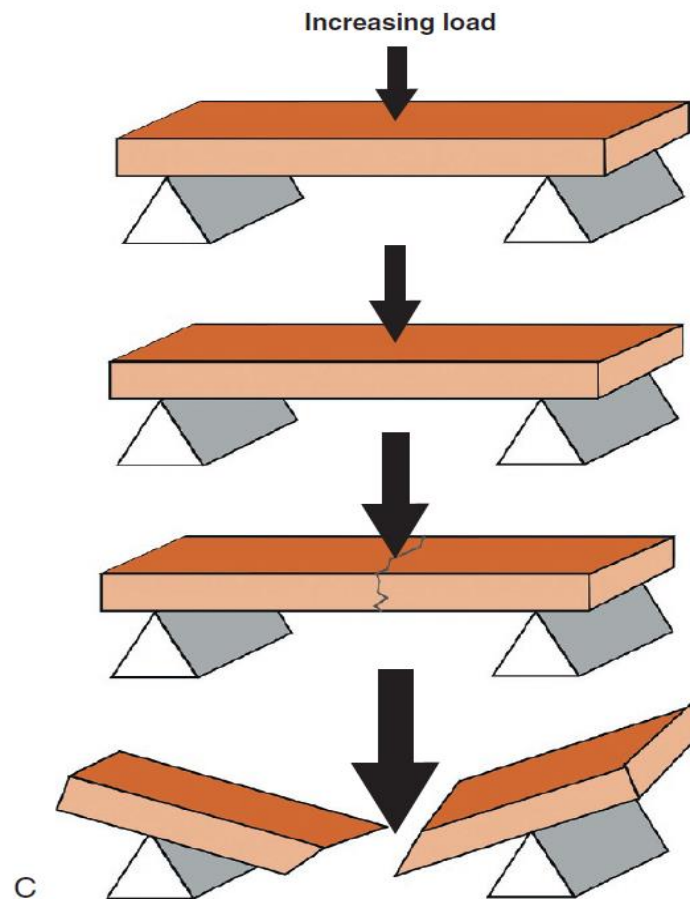
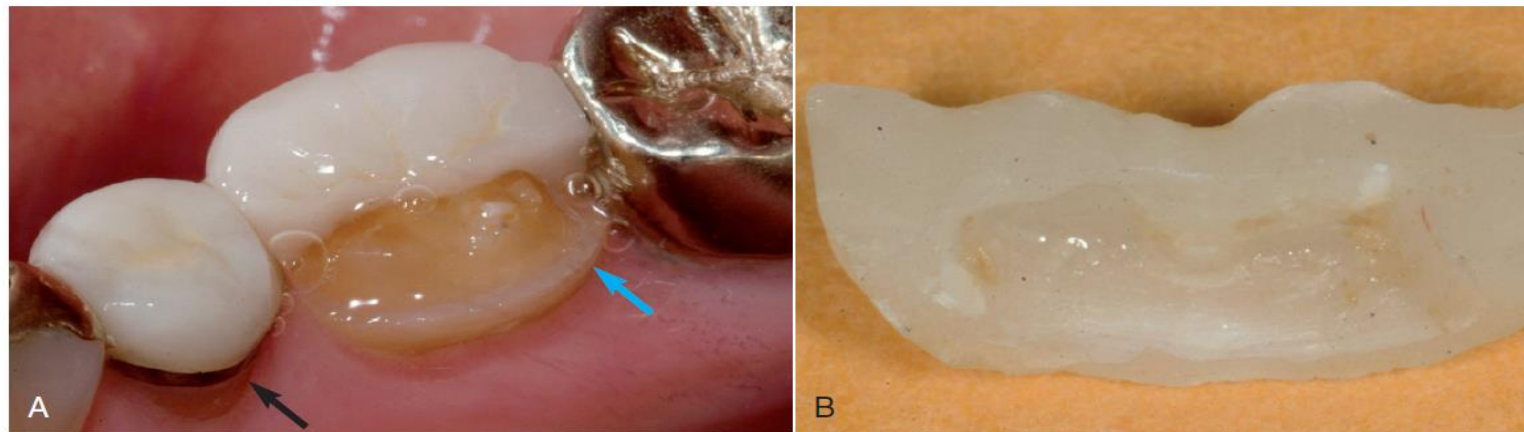
followed by

recently introduced translucent cubic zirconia ceramics (600 to 700 MPa),

Then lithium disilicate– reinforced ceramics (262 to 420 MPa).

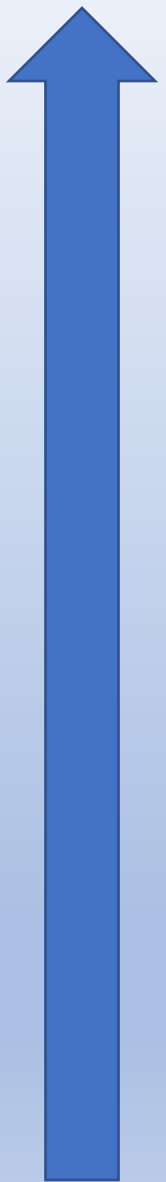
Then flexural strength of leucite-reinforced ceramics is around 120 MPa

-Finally , Flexural strengths of feldspathic dental ceramics range from 70 to 80 MPa, depending on the type of ceramic.



C-Fracture toughness

(Craig Textbook 2020)

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- Fracture toughness is another important property of ceramics
 - it measures the resistance to brittle fracture when a crack is present

- 3Y-TZP ceramics have the highest fracture toughness of all-ceramic materials,

- Followed by lithium disilicate–reinforced ceramics

Then Leucite-reinforced ceramics exhibit slightly higher fracture toughness values than feldspathic porcelain

-The fracture toughness of conventional feldspathic porcelains is the lowest

D- Hardness

The hardness of most glassy or glass-dominated ceramics is substantially higher ($>460 \text{ kg/mm}^2$) than human enamel (343 kg/mm^2), SO $\longrightarrow$ WHEN ceramics occlude against natural teeth, the enamel may wear.

Ceramic-induced wear of opposing enamel may be a problem for fixed prosthetic appliances



Newer crystalline ceramics have a hardness of 1200 to 1500 kg/mm^2 , so the risk of wear of natural teeth is higher.



Current evidence suggests that ceramic-induced wear can be minimized if the restoration surface is very smooth.

Therefore, clinical polishing is very important for these ceramic restorations



II-Other important physical properties

1-coefficient of thermal expansion CTE

A ceramic with a high coefficient of thermal expansion expands a large amount when heated and contracts a large amount when cooled.

The coefficient of thermal expansion is important when a ceramic is bonded to an alloy (ceramic-alloy restoration) or ceramic core (all-ceramic restoration).

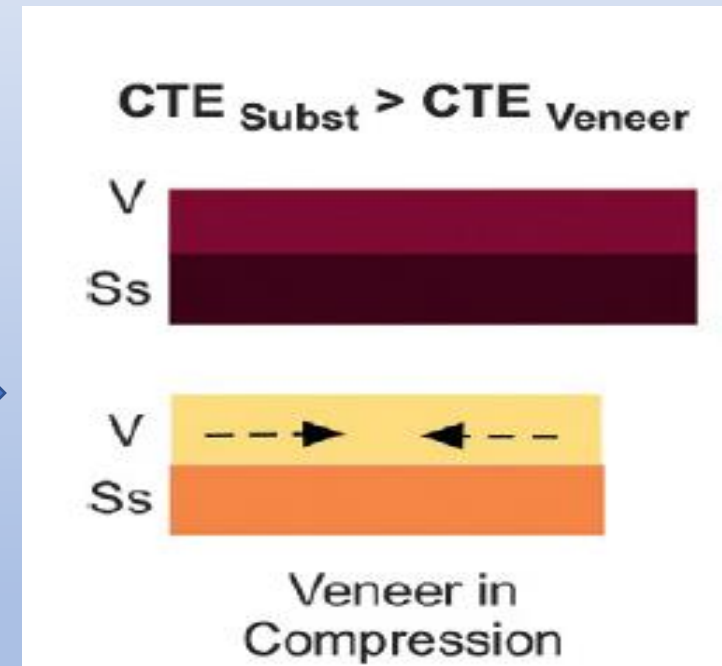
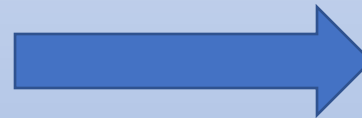
The expansion of the alloy or core and veneering ceramic must be appropriately matched, or the ceramic will fracture when the restoration cools from firing temperature to room temperature

Thermal compatibility of a substructure and veneering ceramic is critical to the clinical success of all ceramic or ceramic–alloy restorations

the CTE of the substructure (as alloy for example) should be slightly greater than that of the veneer.

In this case, the substructure will contract more as the restoration cools to room temperature.

As the veneer is bonded to the substructure, the contraction of the substructure put ceramic veneer under **compression**

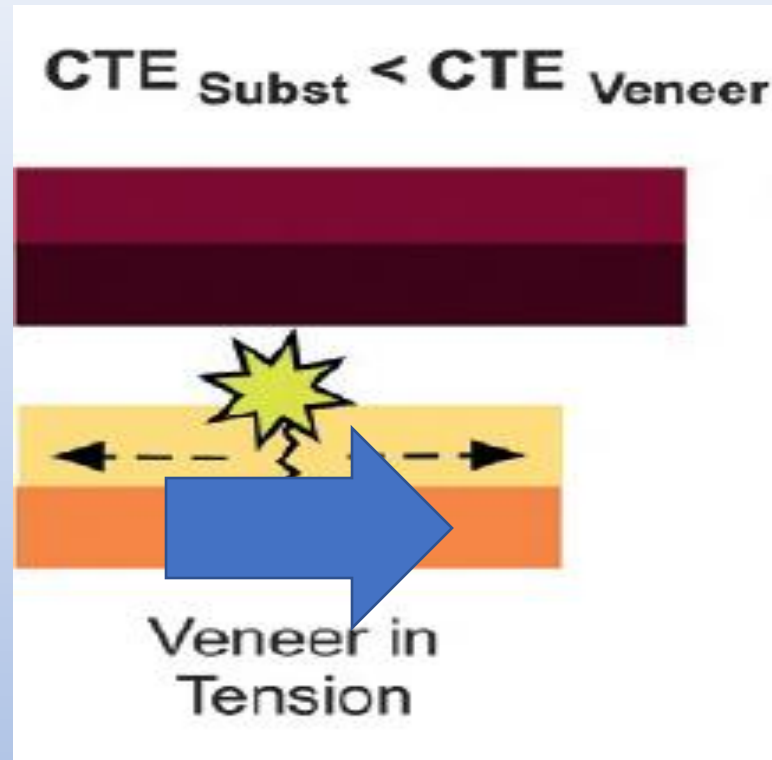


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If the CTE of the veneer is greater than that of the substructure, then the ceramic will tend to be in **tension at room temperature** when bonded to the substructure.

Because ceramics are relatively weak in tension (brittle),

fracture of the ceramic may occur, either immediately or even years after restoration placement.



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2-Other Thermal properties

Glassy ceramics are classified according to their fusing temperature (also called sintering temperature) as follows: 1288° to 1371° C are high fusing; 1093° to 1260° C are medium fusing; and 871° to 1066° C are low fusing.

High-fusing types are used to make denture teeth

medium-fusing types are used to fabricate some all-ceramic restorations

low-fusing types are used for veneering ceramic–alloy and all-ceramic restorations.



For ceramic veneers, a fusing temperature must be selected that the alloy or ceramic substructure can withstand

without distortion or degradation.

The low thermal conductivity
of zirconia-based ceramics was responsible
for early problems of delamination of the
veneering porcelain
WHY????????????????



when fast heating and cooling rates were used.

How to overcome this problem???????
by strict recommendations
for firing of veneering
porcelains on zirconia, including slower heating
and
cooling rates.

(Craig Textbook 2020)



Grade 1 Chipping of a zirconia full-coverage crown



Grade 2 chipping of a zirconia full-coverage crown

3-OPTICAL PROPERTIES OF DENTAL CERAMICS (Craig Textbook 2020)

1-The cementing medium is an important factor in the final appearance of an all-ceramic restoration.

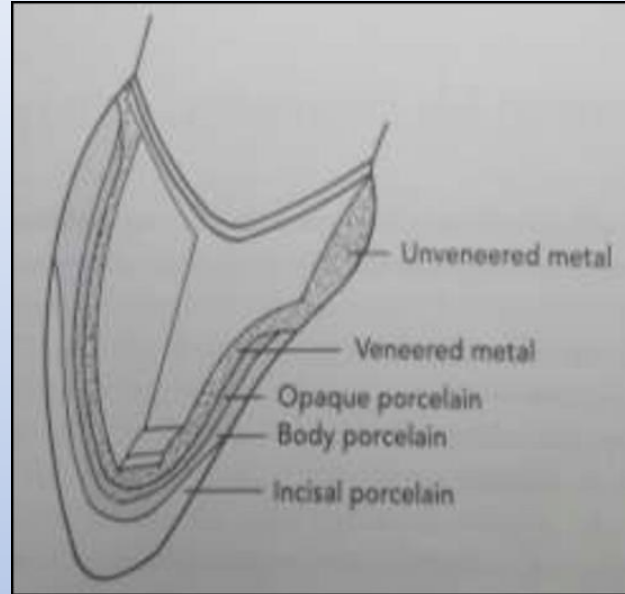
Zirconia based
With their opacity may permit cementation with various luting agents without the need for specific shade matching



more translucent all-ceramic restorations such as leucite-reinforced heat-pressed or machined restorations or lithium disilicate usually require the use of color-matched resin luting agents that are available in different shades.

2-The translucency of opaque, dentin (body), and enamel (incisal) porcelains differs considerably

opaque porcelains have very low translucency, allowing them to efficiently mask metal substructure surfaces. Tin oxide (SnO_2) is an important opacifying oxides for opaque porcelains



https://www.researchgate.net/figure/Layers-of-metal-porcelain-restorations-in-general-7-Metal-Coping-Metal-coping-is-a_fig2_317712331

Enamel porcelains have the highest values of translucency, ranging between 45% and 50%.

Dentin porcelain translucency values range between 18% and 38%

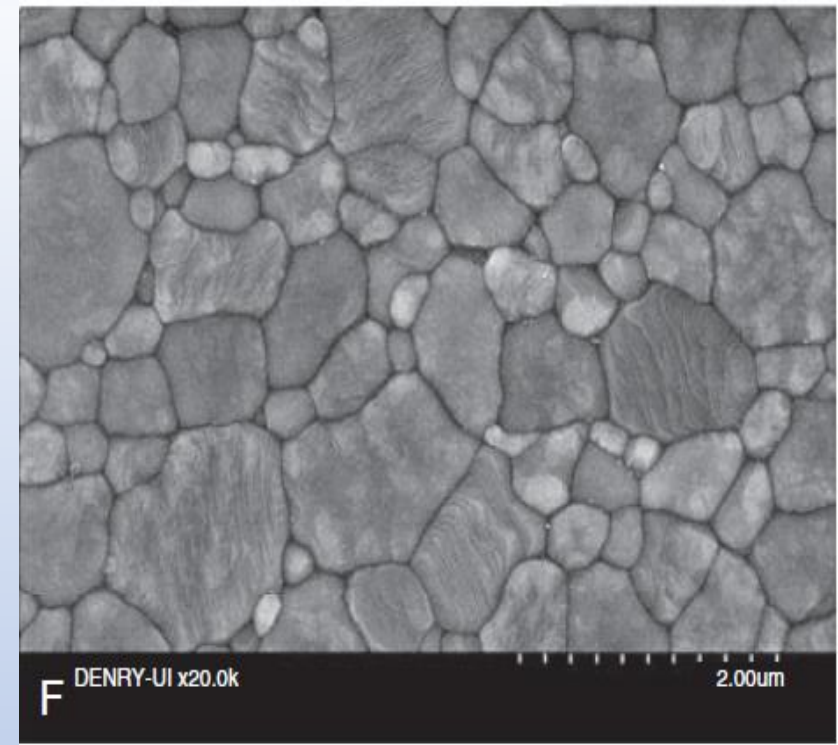
3- Recently,
zirconia ceramics with increased translucency have
become available.

Greater translucency is obtained
by either:

1- decreasing the amount of alumina additive
in the 3Y-TZP composition

or

2- using higher amounts
of yttrium oxide stabilizer to increase the amount of
cubic phase that has greater translucency.



soft-machined sintered high-
translucency zirconia ceramic
(cubic and tetragonal).

III-Biological properties

- Some mass loss (corrosion) of glassy ceramics occurs during service, although the amount of mass released is low relative to polymers or alloys that are used in restorative dentistry.
- There is debate in literature and between researchers about radioactivity hazards of zirconia
- Acidulated phosphate fluoride (APF), one of the most commonly used fluoride gels for caries control, is known to etch glass by selective leaching of sodium ions, thereby disrupting the silica network.
- Dentists should be aware of these long-term clinical effects of fluorides on ceramic and composite restorations and avoid the use of APF gels when composites and ceramics are present

Requirements for a Metal-Ceramic System (Craig Textbook 2020)

- 1-The alloy must have a high melting temperature. The melting range must be substantially higher (greater than 100°C) than the firing temperature of the veneering porcelain and solders used to join segments of an FDP. (metal)
2. The veneering porcelain must have a low fusing temperature so that no creep, sag, or distortion of the framework takes place during sintering. (ceramic)
3. The porcelain must wet the alloy when applied as a slurry to prevent voids forming at the metal-ceramic interface. In general, the contact angle should be 60 degrees or less. (ceramic)
4. A strong bond between the ceramic and metal is essential and is achieved by: 1-chemical reaction of the opaque porcelain with metal oxides on the surface of metal and by 2-mechanical interlocking made possible by roughening of the metal coping. (ceramo-metallic)

5. CTEs of the porcelain and metal must be compatible so that the veneering porcelain **never undergoes** tensile stresses, which would lead to cracking. Metal-ceramic systems are therefore designed so that the CTE of the metal **is slightly higher than** that of the porcelain, thus placing the veneering porcelain **in compression**. (ceramo-metallic)
6. **Adequate stiffness and strength** of the metal framework are especially important for FDPs and posterior crowns. (metal)
7. **High resistance to deformation** at high temperature is essential. Metal copings are relatively thin (0.4 to 0.5 mm); no distortion should occur during firing of the porcelain. (metal)



8. Adequate design of the restoration is critical. The preparation should provide : 1-adequate thickness of the metal coping, 2- adequate thickness of the porcelain. (preparation)

9-During preparation of the metal framework before porcelain application, it is important that all sharp angles be eliminated and rounded to avoid stress concentration in the porcelain.(preparation)

10- If full porcelain coverage is not used and a metal occlusal surface is used, the position of the metal ceramic junction should be located at least 1.5 mm from all centric occlusal contacts. (occlusion)





Machinable All-Ceramic Materials

A-Ceramic materials can be machined in the **fully sintered state** and termed ***hard machining*** and restorations produced in this way are machined directly to final size. So, **in hard machining ceramics are fully sintered before machining.**

B-Some all-ceramic materials can also be machined in a **soft, partially sintered** state then **later fully sintered**. This is called ***soft machining***.

The latter technique requires milling of enlarged restorations to compensate for sintering shrinkage as in zirconia

Hard machining of lithium disilicate ceramics

lithium disilicate glass-ceramic blocks are available for hard machining by CAD/ CAM.

The lithium disilicate glass-ceramic blocks are machined in a partially crystallized state, which corresponds to a softer and easier-to-machine state.

After the restorations are machined, a heat treatment at 850°C for 10 minutes is performed to complete the crystallization process, and therefore ensure that the lithium disilicate ceramic possesses improved mechanical properties before glazing and cementation.

(Craig Textbook 2020)



Fig 12-1

References

Figures and text information :

1- Dental materials foundations and applications 11th Edition

2- Craig's restorative dental materials 14th Edition.

THANK YOU